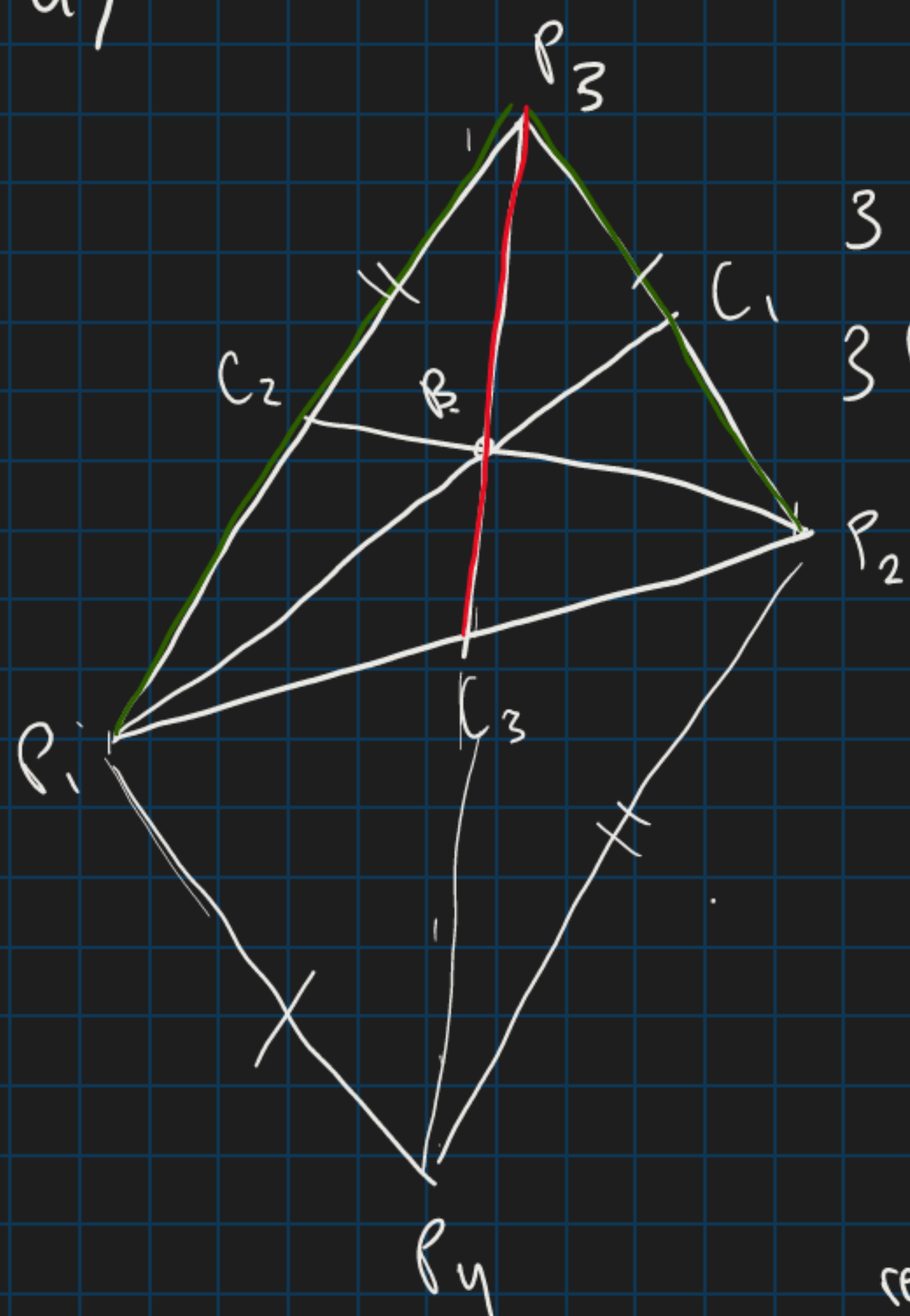


5. a)



$$3 \vec{P_3 B} = 1(\vec{P_3 P_1} + \vec{P_3 P_2})$$

$$3 \vec{P_3 B} = 2 \vec{P_3 C_3}$$

$$\vec{P_1 P_4} = \vec{P_3 P_2}$$

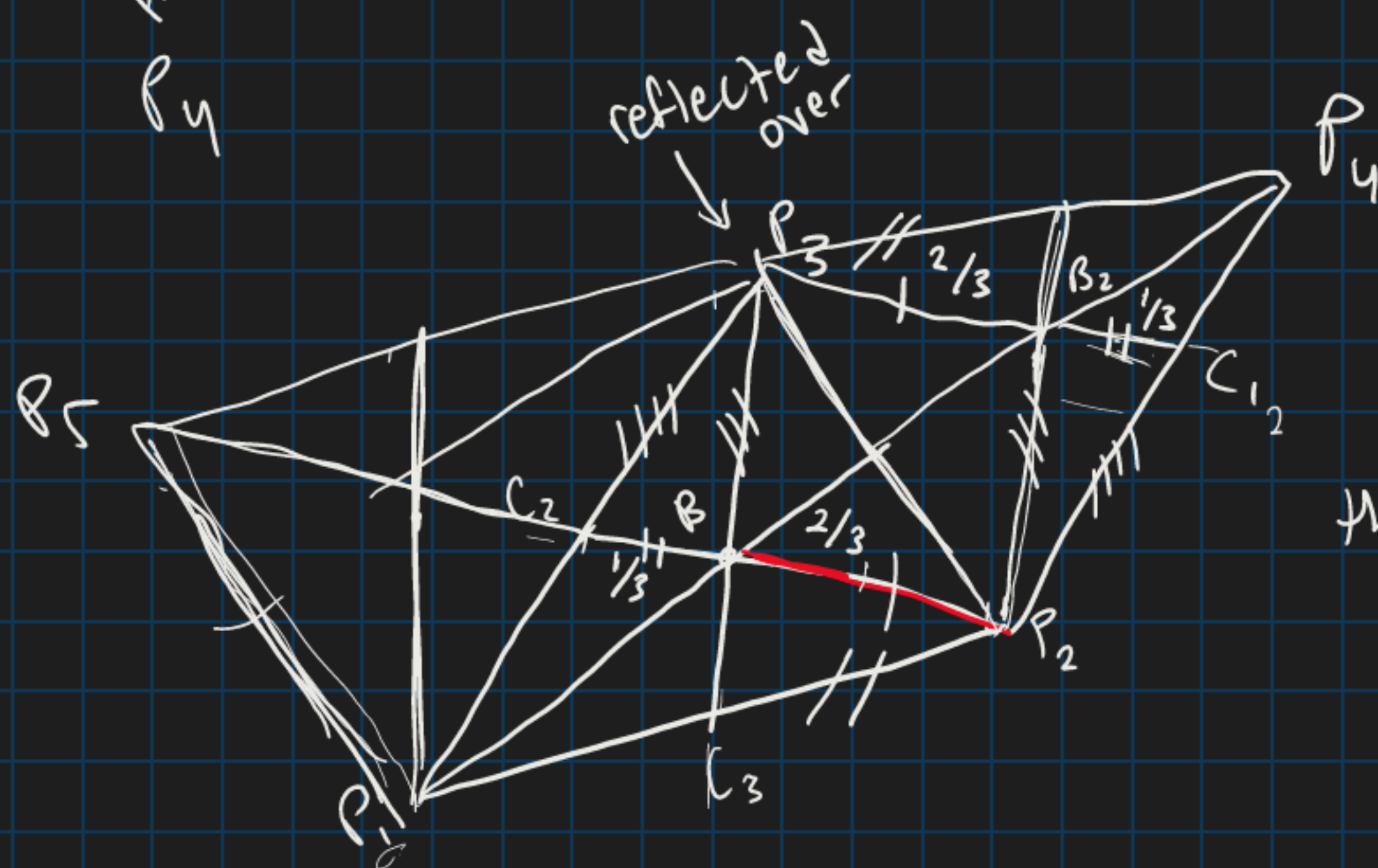
$$\vec{P_3 P_1} + \vec{P_3 P_2} = 2 \vec{P_3 C_3}$$

$$\vec{P_1 P_4} + \vec{P_3 P_1} = \vec{P_3 P_4} \quad \vec{P_3 P_4} = 2 \vec{P_3 C_3}$$

Quadrilateral $P_1 P_3 P_2 P_4$ is a parallelogram (formed out of reflecting $\triangle P_1 P_3 P_2$ over $P_1 P_2$ to make 2 congruent $\triangle$ s, sharing side $P_1 P_2$)

$$\vec{P_3 P_2} + \vec{P_3 P_1} = 2 \vec{P_3 C_3}$$

b)



$$\vec{P_2 B} = \vec{B P_3}$$

$$\vec{P_2 C_2} = \vec{C_1 P_3}$$

$$\text{therefore } \vec{P_2 B} = \frac{2}{3} \vec{P_2 C_2}$$

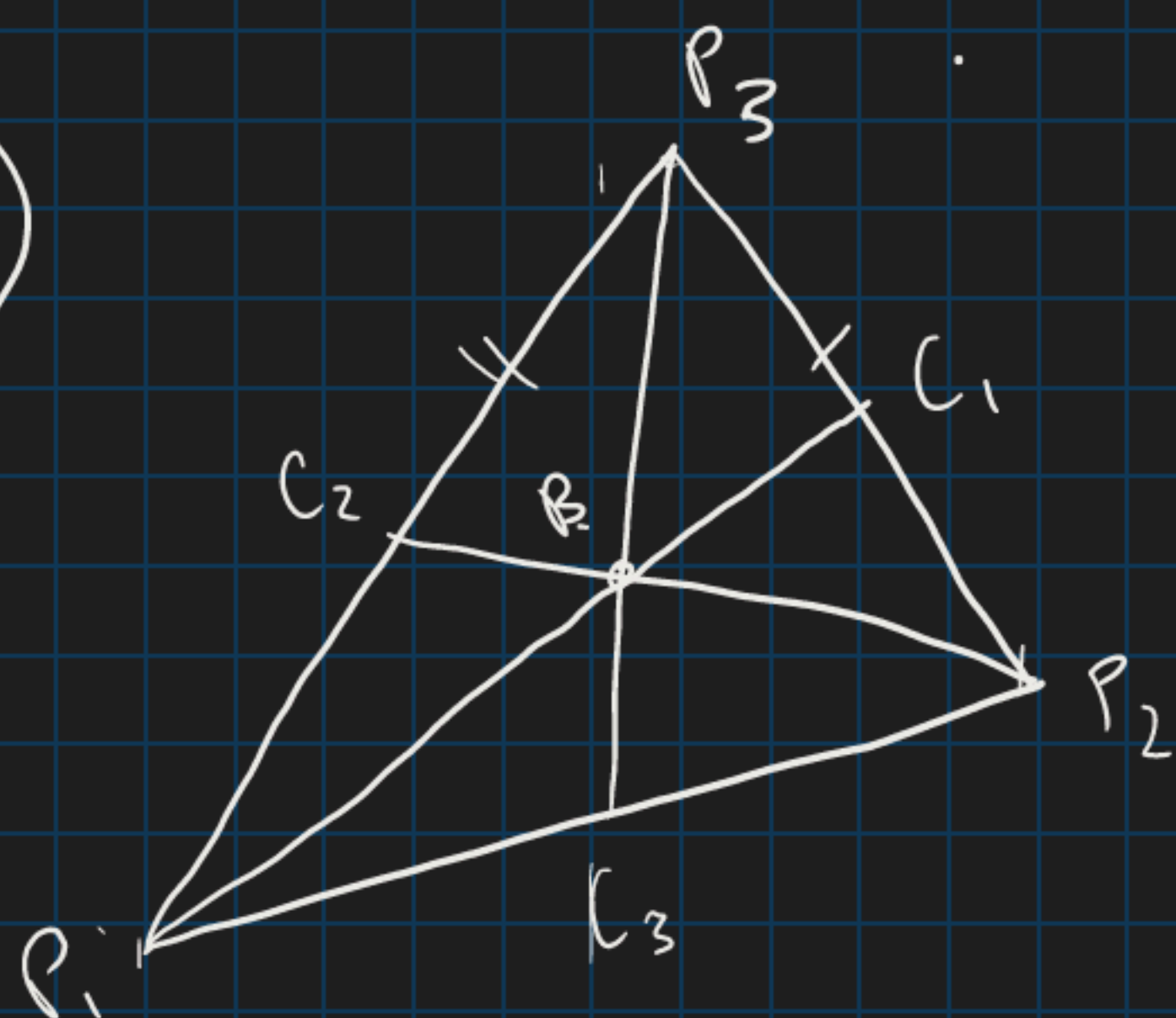
$$\Rightarrow \vec{P_2 B} = \frac{2}{3} \vec{P_2 C_2}$$

$$\vec{P_2 P_1} + \vec{P_1 P_5} = \vec{P_2 P_1} + \vec{P_2 P_3} = \vec{P_2 P_5}$$

$$3 \times \vec{P_2 B} = \vec{P_2 P_1} + \vec{P_2 P_3}$$

$\triangle P_1 P_2 P_3$ is reflected over $P_1 P_3$ to form a parallelogram of 2 equal triangles. $P_1 P_2$ and $P_3 P_5$ are thus parallel. Since the triangle was reflected, $B P_3$ and $P_2 B_2$ are also parallel and of equal length, thus $B P_3 B_2 P_2$ is a parallelogram. $P_2 B$ and $B_2 P_3$ are then congruent, and since $P_2 C_2$ is equal to $C_1 P_3$ (shown by parallelogram $C_2 P_3 C_1 P_2$). Assuming part a, $\vec{P_3 B} = \frac{2}{3} \vec{P_3 C_3}$ is true, then $P_2 B$ is $\frac{2}{3}$ of $P_2 C_2$. $\vec{P_2 P_1} + \vec{P_1 P_5} = \vec{P_2 P_1} + \vec{P_2 P_3} = \vec{P_2 P_5}$. $P_2 B$ is $\frac{2}{3}$ of $\frac{1}{2}$ of $P_2 P_5$, meaning $P_2 B$ is $\frac{1}{3}$ of $\vec{P_2 P_1} + \vec{P_2 P_3}$.

c)



$$\vec{B P_1} + \vec{B P_2} + \vec{B P_3} = 0 \quad (\because B \text{ is the triangle's centroid})$$

$$\vec{P_2 B} + \vec{P_1 B} + \vec{P_3 B} = \frac{1}{3}(\vec{P_2 P_1} + \vec{P_2 P_3}) + \frac{1}{3}(\vec{P_3 P_1} + \vec{P_3 P_2}) + \frac{1}{3}(\vec{P_1 P_3} + \vec{P_1 P_2})$$

$$\vec{P_2 B} + \vec{P_1 B} + \vec{P_3 B} = \frac{1}{3}(\cancel{\vec{P_2 P_1}} + \cancel{\vec{P_2 P_3}} + \cancel{\vec{P_3 P_1}} + \cancel{\vec{P_3 P_2}} + \cancel{\vec{P_1 P_3}} + \cancel{\vec{P_1 P_2}})$$

$$\vec{P_2 B} + \vec{P_1 B} + \vec{P_3 B} = 0$$

$$\vec{P_2 B} + \vec{P_1 B} + \vec{P_3 B} = -(\vec{B P_2} + \vec{B P_1} + \vec{B P_3})$$

$$\vec{B P_1} + \vec{B P_2} + \vec{B P_3} = 0$$